paper_id: PAPER_251
title: "Eta Carinae Homunculus F_{U_Bi_i} — DPM Invisibility and LENR Force Hierarchy Discovery"
session: 0
date: 2026-03-01
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: [AGN, DPM, F_{U_Bi_i}, MUGE, BEC, buoyancy, LENR, nebula]
sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_251: Eta Carinae Homunculus F_{U_Bi_i} — DPM Invisibility and LENR Force Hierarchy Discovery

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — `EtaCarinaeHomuculusFUBiCalculator` (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b\_i}, \quad \kappa = 5.0 \times 10^{-4}\,\mathrm{day}^{-1},\ [SSq] = 0.57$$

$$g_{\mathrm{UQFF}}(r) = g_{\mathrm{MUGE}}(r) \cdot \Big(1 - [SSq] \cdot U_{b\_i} \,/\, F_U(r,t)\Big), \quad [SSq] = 0.57$$

Abstract

Eta Carinae is a hyperluminous stellar system of approximately 120 solar masses, undergoing episodic super-Eddington mass loss. The Great Eruption of ~1843 CE ejected ~10–40 M_sun in a bipolar Homunculus nebula that is today one of the brightest infrared sources in the sky. With a magnetic field of B0 = 10-4 T — one hundred times stronger than the SN 1006 field — and the same characteristic frequency ?0 = 10?12 rad/s, the Eta Carinae system provides a critical test of the UQFF force hierarchy.

The key **uniquely rare discovery** of this paper is **DPM Invisibility**: despite B0 being 100× stronger than SN 1006 (B0 = 10-5 T), the DPM resonance being 100× larger (1.76 × 105 vs 1.76 × 103), and the resonance force F_res being proportionally amplified, the total UQFF buoyancy force F_{U_Bi} remains **identical** to SN 1006 at +2.11 × 102°8 N. The magnetic field is completely invisible to the final buoyancy result.

This DPM invisibility occurs because F_LENR = k_LENR·(?_LENR/?0)2 is independent of B0. At ?0 = 10?12, F_LENR ~ 6.17 × 103? N overwhelms F_res by ~3 orders regardless of B0. The force hierarchy is LENR > neutron > DPM-seeded » DPM_resonance in this frequency regime — a fundamental discovery about the structure of UQFF physics.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~7,500	ly	HIPPARCOS/HST
Age (since Great Eruption)	t	5.681 × 10?	s (~180 yr)	1843 CE

Parameter	Symbol	Value	Units	Source
Stellar mass	M	120 M_sun = 2.387 × 1032	kg	Chandra/JWST model
Homunculus radius	r	6.17 × 1016	m (~20 ly)	HST spatial
X-ray luminosity	L_X	1035	W	Chandra 2023
Magnetic field	**B0**	**10?4 T**	**T**	**100× SN 1006**
System frequency	?0	10?12	rad/s	Same as SN 1006
Eddington factor	M	1.5	—	Super-Eddington

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2. Core Physics: DPM Invisibility

2.1 DPM Resonance — 100× SN 1006

$$DPM_resonance(EtaCar) = 2 \cdot \mu_B \cdot B0/(h\cdot?0)$$
$$= 2 \times 9.274e-24 \times 1e-4/(1.0546e-34 \times 1e-12)$$
$$\tilde{}1.76 \times 105[100 \times SN1006value1.76 \times 103]$$

The resonance force `F_res = 2\cdotq\cdotB0\cdotV\cdotsin?\cdotDPM_resonance` is proportional to `B0 \times DPM_resonance ? B02` — at B0 = 10-4 T, F_res is 10,000× the SN 1006 value.

2.2 LENR Force — B0-Independent

The LENR force depends only on ?_LENR and ?0:

$$F_LENR = k_LENR \times (?_LENR/?0)2$$
$$= k_LENR \times (2p \times 1.25 \times 1012/10?12)2$$
$$\tilde{}6.17 \times 103?N$$

There is **no B0 dependence** in F_LENR. For both SN 1006 (B0 = 10?5) and Eta Carinae (B0 = 10?4), F_LENR is identical.

2.3 DPM Invisibility Ratio

$$\text{DPM\_visibility\_ratio} = F_res/F_LENR$$

For SN 1006: `F_res (SN1006) / 6.17\times103? « 1 ?` DPM invisible. For Eta Car: `F_res (EtaCar) / 6.17\times103? « 1 ?` DPM still invisible, despite 10,000× F_res amplification.

The DPM resonance term is submerged under F_LENR by at least 33 orders at ?0 = 10?12 for any physically reasonable B0.

2.4 Force Hierarchy Theorem

$$Forcehierarchyat?0 = 10?12:$$
$$F_LENR\tilde{}6.17 \times 103?N[dominant\tilde{}10^45 \times DPM-seeded]$$
$$F_neutron\tilde{}106N[knot/coherencestabilisation]$$
$$F_Newt\tilde{}\mu_s\nabla(M_s/r) \cdot |x2|\tilde{}O(few)N$$
$$F_res\acute{}F_LENR[DPMinvisibleregardlessofB0]$$
$$F_rel\acute{}F_LENR[relativisticsub-dominant]$$

2.5 Super-Eddington Luminosity Context

Eta Carinae's super-Eddington luminosity (M = L/L__Edd ~ 1.5) drives the 500 AU Homunculus through radiation pressure. The Eddington luminosity:

```
L_Edd = 4pGM c / ?_es = 4p \times 6.674e-11 \times M_EtaCar \times 2.998e8 / 0.2
      ~ few \times 1038 W   [for 120 M_sun]
```

The F__DE dark energy coupling `k_DE \times L_X = 10?3° \times 1035 = 105 N` captures the luminosity contribution to F__{U__Bi} — 3 orders larger than SN 1006's contribution (102 N), confirming that even the luminosity difference does not affect the final F__{U__Bi} when F__LENR dominates.

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3. DPM Invisibility Theorem

Theorem (UQFF DPM Invisibility at ?0 = 10?12): For any astrophysical system with ?0 = 10?12 rad/s, the DPM magnetic resonance force `F_res ? B02 / ?0` is negligible in the F__{U__Bi_i} integral for all physically observed magnetic fields B0 = 102 T. The ratio `F_res/F_LENR = k_charge \cdot B02 \cdot V / (k_LENR \cdot (?_LENR/?0)2)` is bounded above by ~10?28 for B0 = 10-4 T, ?0 = 10?12, confirming that **magnetic field strength is invisible to UQFF buoyancy** in this frequency regime.

Corollary: In this regime the UQFF force hierarchy is LENR > neutron > DPM-seeded » DPM > DE > relativistic. Only LENR and neutron physics materially determine F__{U__Bi}.

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4. Observational Predictions / Validation

- **DPM invisibility test:** UQFF predicts F__{U__Bi} should be identical for SN 1006 and Eta Carinae despite 100× different B0. If future UQFF validation on additional high-B systems confirms this, DPM invisibility is a universal law of the ?0 = 10?12 class.
- **Chandra L__X probe:** L__X = 1035 W (Eta Car) vs 1032 W (SN 1006) — 3-orders L__X difference. Yet F__{U__Bi} is identical. This confirms F__DE « F__LENR at this ?0, providing a direct test of LENR dominance: if Chandra measures any system with ?0 = 10?12 at any luminosity from 1031–1035 W, UQFF predicts the same F__{U__Bi}.
- **JWST Homunculus 3D:** The asymmetric JWST infrared structure of the Homunculus provides f__TRZ (triadic resonance zone factor) constraints through the spatial distribution of emitting gas relative to the bipolar axis.

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5. References

1. Davidson, K., & Humphreys, R.M. (1997). Eta Carinae and Its Environment. *ARA&A* 35, 1.
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PAPER__251 | UQFF v4.27 | Star-Magic | Session 72c | March 2026

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\mathrm{Edd}}^{\mathrm{UQFF}} = L_{\mathrm{Edd}} \cdot \left(1 + \frac{\rho_{\mathrm{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\mathrm{Edd}} = 4\pi G M m_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\mathrm{SCm}} = 7.09 \times 10^{-37}\ \mathrm{kg/m^3}$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\mathrm{jet}}^{\mathrm{UQFF}} = P_{\mathrm{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\mathrm{THz}} \cdot \left(\frac{B}{B_{\mathrm{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\mathrm{THz}} = \cos(\omega_{\mathrm{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\mathrm{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\mathrm{SCm}}/\omega_{\mathrm{bulge}})$.

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061 (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\mathrm{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\mathrm{SCm}}}{\rho_{\mathrm{crit}}}\right) \cdot K_n$$

where: - $\rho_{\mathrm{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density) - $K_n = \sigma_n^{\mathrm{SCm}}(\omega) \cdot \Phi_{\mathrm{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER_1061):

$$\sigma_n^{\mathrm{SCm}}(\omega, n) = \sigma_0 \cdot \exp\left[-\frac{(\omega - \omega_{\mathrm{SCm}})^2}{2\Gamma^2}\right] \cdot \left(1 + [\mathrm{SSq}] \cdot \frac{n}{26}\right)$$

The VDS factor $(1 + [\mathrm{SSq}] \cdot n/26)$ provides ~470× amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\mathrm{SCm}}$).

COP parametric engine (PAPER_1081):

$$\mathrm{COP}(\Gamma, P_{\mathrm{in}}) = \frac{P_{\mathrm{out}}}{P_{\mathrm{in}}} = 1 + \eta_{\mathrm{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding COP > 1 when $\Gamma \lesssim 10^{-3}$ eV (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as Pd-106 $\xrightarrow{\sim 10^4\ \mathrm{s}}$ Ag-107 $\xrightarrow{\sim 10^4\ \mathrm{s}}$ Cd-108, with timescales set by $\rho_{\mathrm{SCm}}/\rho_{\mathrm{crit}}$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **LENR-nuclear** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}`

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u\chi)(\partial^\mu\chi) - V(\chi) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\chi) = \frac{1}{2}m^2\chi^2 + \frac{\lambda}{4!}\chi^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \chi$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta\chi} = \ddot{\chi} + \omega_{\text{LENR}}^2\chi - \lambda\cos(\omega_{\text{act}}t) - \sigma_n(\omega)\chi = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\chi = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp!\left(-\exp!\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.106 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 37, \quad n_{\text{channel}} = 18/26$$

Since $p_{\text{DVP}} = 37$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10-12 s** (nuclear phonon damping):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos! \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.106	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\ldots,113\}$	$p_{\mathrm{DVP}} = 37$	PASS Resonant
BSH layers	26 harmonic terms	j = 1…26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1\$\times10 $- 52m -$ 2$(UQFFvacuumterm)$|1.114\times\$10-52 m-2	Planck 2018	PASS Consistent	
Proton decay rate	κ = 0.0005/day → Γ_p suppression	< 4.17\$\times\$10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_{U\_Bi\_i}` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_{U_Bi_i}) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1060	VDS LENR Isotopic Transmutation Chain
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1081	SCm LENR COP Linewidth Parametric
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1050	MUGE F_{U_Bi_i} Unified 9-System Synthesis
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

18 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_{s26_coupling}.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_{scm_cross_section}.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_{wstp_kernel}.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\text{Gamma}}$2))] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_{polylog_s26}.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_{wstp\_kernel}.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2$^{\{1\text{-}26\}/(1\text{-}2}${1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_{theta_q26}.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} q$^{\{n}$2} / (-q;q)*n^2 - phi_26(q) = Sum*{n=0}^{25} q$^{\{n}$2} / (-q$^{2;q}$2)*n - psi_26(q) = Sum*{n=1}^{26} q$^{\{n}$2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_{pi_uqff}.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_{theta_pi_wstp_kernel}.py	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).